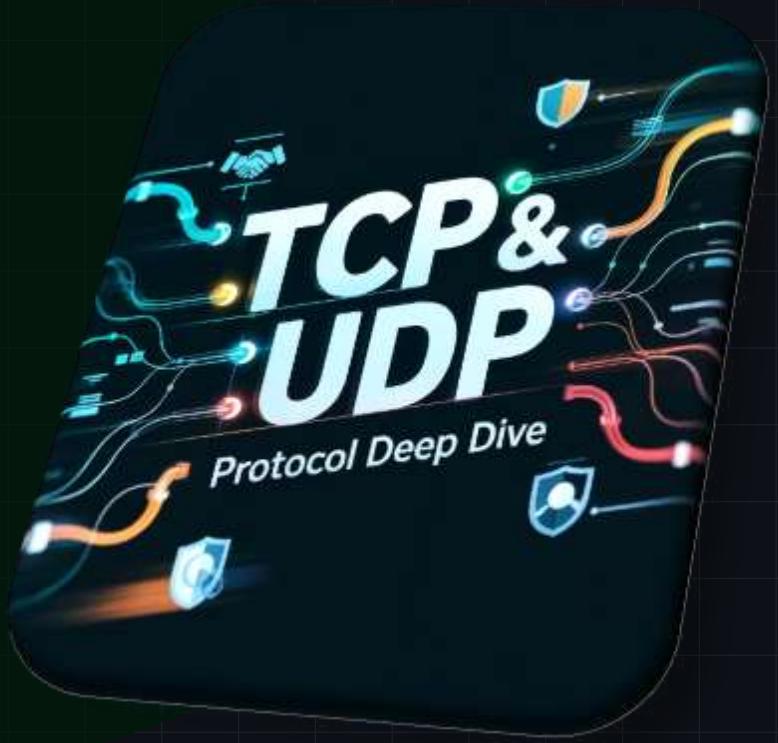
CYBERSECURITY NOTE | NETWORKING

TCP & UDP

Network Protocol Deep Dive

3-Way Handshake · TCP Flags · SYN Flood · UDP Spoofing · Stealth Scans

TCP	UDP	SYN	ACK
FIN	RST	PSH	URG



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TCP — Transmission Control Protocol

WHAT IS TCP?

A Connection-Oriented protocol. Before any data is sent, both sides must establish a formal agreement — the handshake. Every packet is tracked, acknowledged, and retransmitted if lost.

✓ Reliable Delivery

Every packet is acknowledged. Lost packets are retransmitted automatically.

⇄ Ordered Data

Packets arrive in sequence. TCP reassembles them in the correct order.

⌛ Flow Control

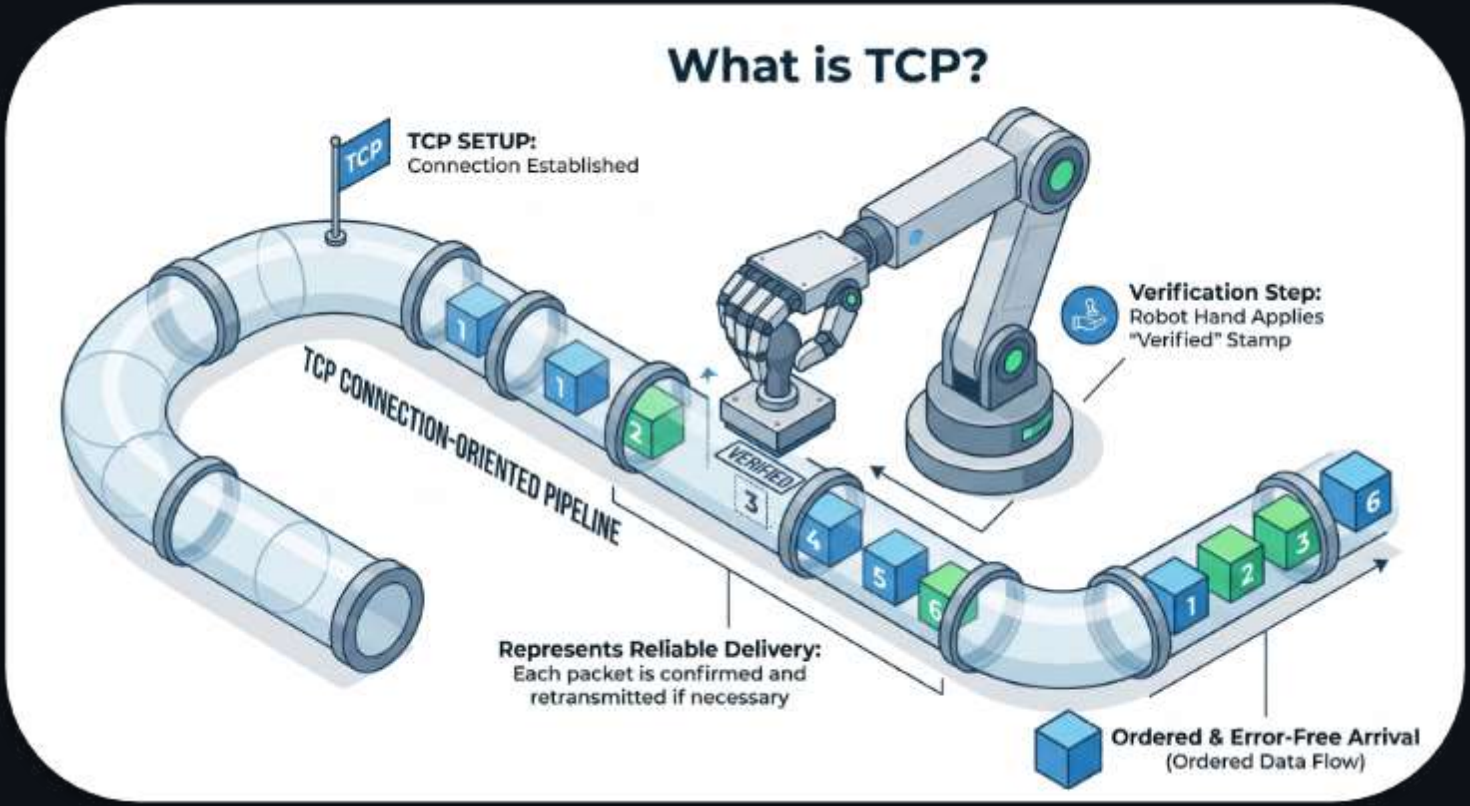
Prevents the sender from overwhelming the receiver using windowing.

TRADEOFF

TCP reliability comes at a cost: overhead. The handshake, acknowledgements, and retransmissions add latency. This is why real-time applications (gaming, video) prefer UDP.

OSI LAYER 4 (Transport Layer) — Port numbers identify services: HTTP=80, HTTPS=443, SSH=22, DNS=53

TCP — Transmission Control Protocol



TCP 3-Way Handshake

The connection establishment protocol — both parties synchronize before data transfer.

CLIENT | **SERVER**

STEP 1
SYN

"Hello! Can I connect? ISN=1000" →

SYN (Seq = 1000)

STEP 2
SYN-ACK

← *"Got 1000 (ACK=1001). My ISN=5000"*

SYN-ACK (Seq=5000, Ack=1001)

STEP 3
ACK

"Got 5000 (ACK=5001). Connection ready!" →

ACK (Ack = 5001)

✓ **CONNECTION ESTABLISHED — Data transfer can begin**

Offensive: SYN Scan vs SYN Flood

STEALTH SCAN (nmap -sS)

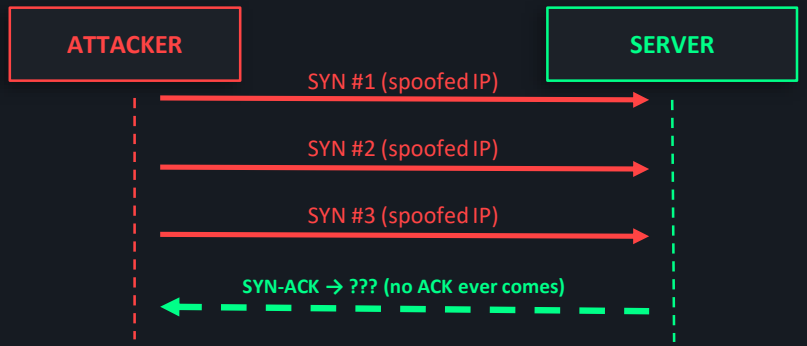
Port Reconnaissance — No full connection

Half-open → Not logged by old firewalls

If SYN-ACK received → Port OPEN
If RST received → Port CLOSED

SYN FLOOD ATTACK (DoS)

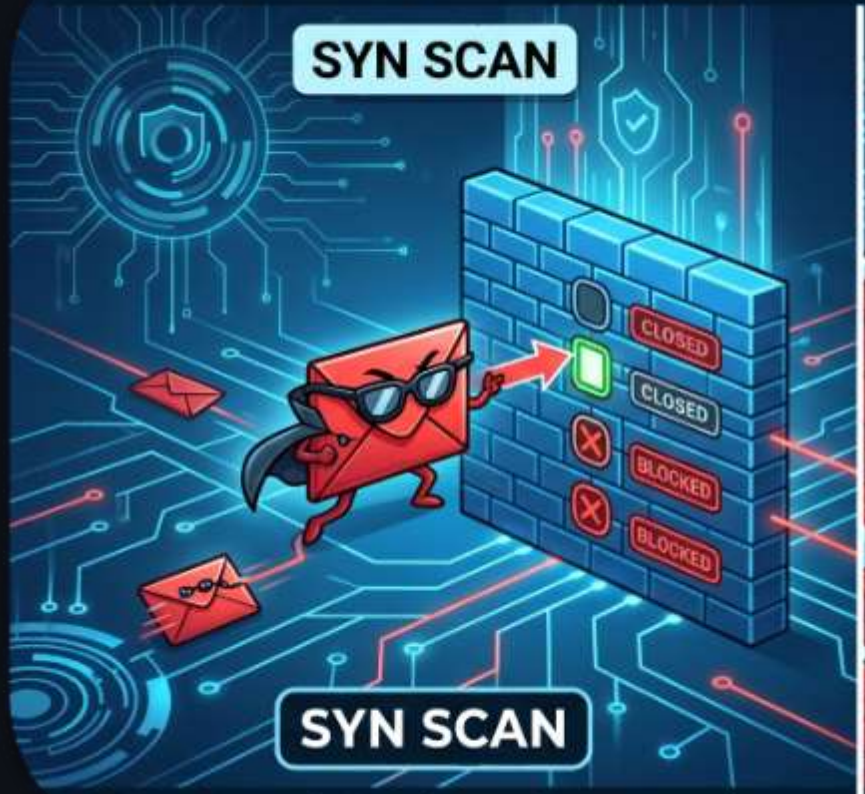
Denial of Service — Server exhaustion



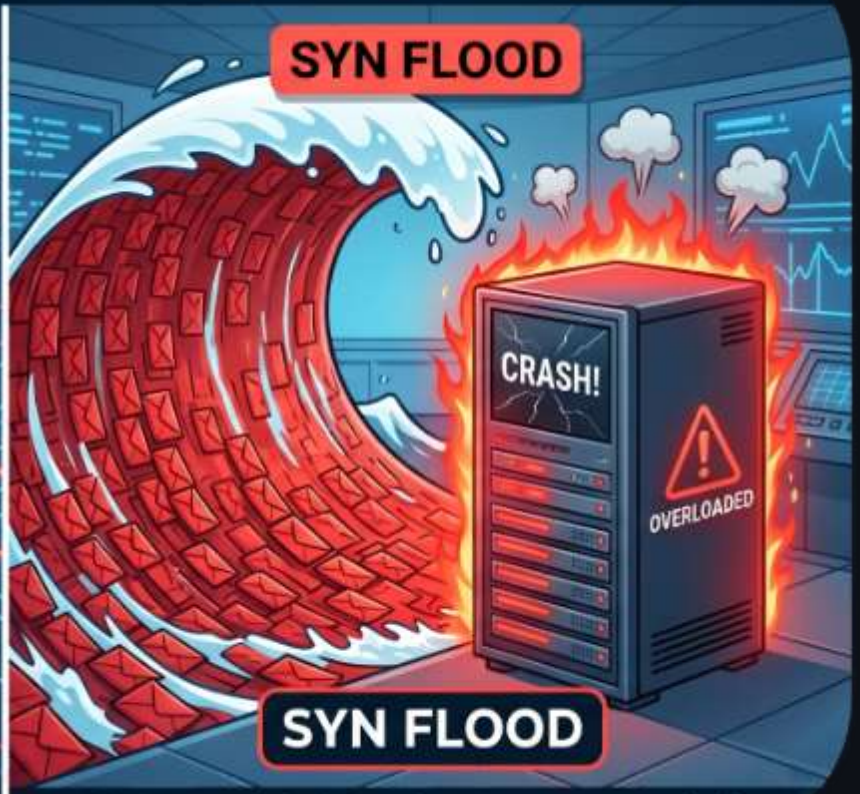
Server RAM fills with half-open connections → CRASH / DoS

FIX: SYN Cookies, Rate Limiting, Firewall rules

OFFENSIVE CYBERSECURITY: SYN SCAN VS SYN FLOOD



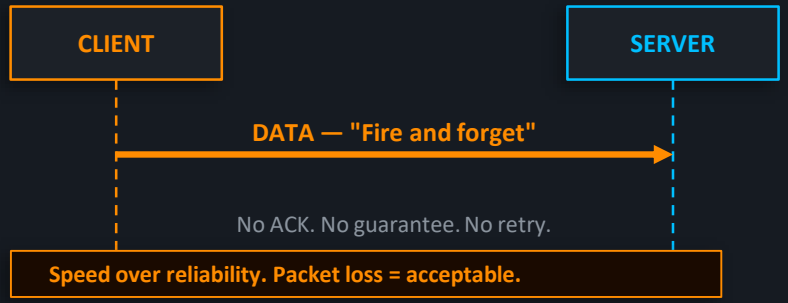
GOAL: Network Mapping and Port Detection.



GOAL: Denial of Service, Resource Depletion.

UDP — User Datagram Protocol

UDP — NO HANDSHAKE



Speed over reliability. Packet loss = acceptable.

USE CASES (Speed > Reliability)

- 🎮 **Online Gaming** — 1 dropped frame OK
- 📹 **Video Streaming** — Slight glitch > pause
- 🌐 **DNS Lookups** — Fast single query
- 📡 **VoIP Calls** — Real-time audio

TCP vs UDP

Feature	TCP	UDP
Connection	Required (3-way)	None
Reliability	Guaranteed	Not guaranteed
Order	Maintained	Not maintained
Speed	Slower	Faster
Error Check	Full	Minimal
Use case	HTTP, SSH, FTP	DNS, VoIP, Games
Attack vector	SYN Flood	UDP Flood / Spoof

UDP — User Datagram Protocol

TCP Flag Family

6 flags in the TCP header — each controls connection state. Misuse = attack.

SYN

Synchronize

Initiates a connection. The opening move.

⚠ SYN Flood DoS

RST

Reset

Abrupt termination. Error or rejection.

⚠ RST Injection

ACK

Acknowledge

Confirms receipt. Used in almost every packet.

⚠ ACK Flood

PSH

Push

Send data to app immediately, skip buffer.

⚠ PSH+URG combo scans

FIN

Finish

Graceful connection close — 'I am done sending.'

⚠ FIN Scan (nmap -sF)

URG

Urgent

Prioritize this data — process before queue.

⚠ URG pointer exploits

SLIDE 6: TCP FLAG FAMILY
ATTACK
ATTACK
ATTACK
off
off
on
off
off
SYN
ACK
FIN
RST
PSH
URG

TCP 4-Way Termination

Graceful connection teardown — both sides independently close their half of the connection.

CLIENT | **SERVER**

STEP 1 — **FIN**

FIN (Client done sending) →

STEP 2 — **ACK**

← ACK (Server received FIN)

STEP 3 — **FIN**

← FIN (Server also done)

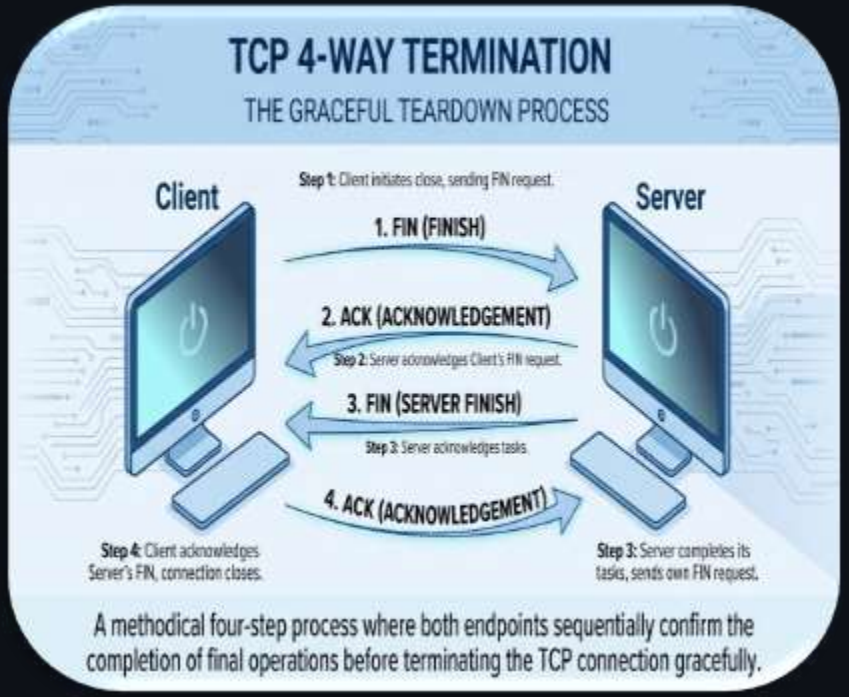
STEP 4 — **ACK**

ACK (Client confirms, goodbye) →

Why 4 not 3?

Half-close allowed

✓ CONNECTION TERMINATED — Both sides closed gracefully



Advanced Nmap Scans

Exploiting TCP flag logic to bypass firewalls and evade detection.

Xmas Scan (nmap -sX)

FLAGS

FIN + PSH + URG

HOW IT WORKS

All 3 flags lit simultaneously — looks like a 'Christmas tree'

OPEN PORT

No response (port open or filtered)

CLOSED PORT

RST packet received

BYPASS

Bypasses stateless firewalls and old Linux systems

FIN Scan (nmap -sF)

FLAGS

FIN only

HOW IT WORKS

Sends just a FIN — looks like the end of a conversation that never started

OPEN PORT

No response (RFC 793 compliant open ports ignore it)

CLOSED PORT

RST packet received

BYPASS

Evades firewalls watching for SYN packets

NULL Scan (nmap -sN)

FLAGS

No flags set

HOW IT WORKS

Sends a packet with zero flags — completely abnormal in real traffic

OPEN PORT

No response

CLOSED PORT

RST packet received

BYPASS

Useful against non-RFC-compliant stacks (Windows may not respond predictably)

Advanced Nmap Scans

Purple Team: Detection Engineering

If you can attack it, you must know how to detect and defend it.

SYN Flood

SIGNAL

High volume SYN packets with no corresponding ACK

DETECTION RULE / INDICATOR

```
alert tcp any any -> $HOME_NET any (flags:S; detection_filter: track by_dst,
count 100, seconds 1;)
```

FIX: SYN Cookies, Rate limiting, Firewall SYN threshold

Xmas / FIN / NULL Scan

SIGNAL

TCP packets with illegal flag combinations (FIN+PSH+URG, or no flags)

DETECTION RULE / INDICATOR

```
alert tcp any any -> $HOME_NET any (flags:FPU; msg:"XMAS Scan Detected";)
```

FIX: IDS signature rules, stateful firewall inspection

UDP Flood / Spoof

SIGNAL

High UDP traffic volume to random ports; source IPs changing rapidly

DETECTION RULE / INDICATOR

```
Monitor for ICMP Unreachable storm (response to closed UDP ports)
```

FIX: Rate limit UDP per source, ingress filtering (BCP38), Anycast diffusion

Tools: Wireshark · Snort / Suricata · Zeek · Elastic SIEM · tcpdump

Purple Team: Detection Engineering

Key Takeaways

"The handshake is not just a protocol — it is the attack surface."

01

TCP is reliable but attackable

3-way handshake = connection. Missing ACK = SYN Flood.

02

UDP trades safety for speed

No handshake = easy spoofing. UDP Flood = DoS.

03

TCP Flags are your attack surface

Abnormal flag combos = Xmas, FIN, NULL scans.

04

4-Way teardown = graceful exit

FIN+ACK+FIN+ACK. Half-close is legal.

05

Detect what you can attack

SYN w/o ACK = flood. Flags combo = scan. Rate spike = UDP flood.

#Cybersecurity #TCP #UDP #NetworkSecurity #PurpleTeam

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SUMMARY & TAKEAWAYS

KEY TAKEAWAYS

1 INNOVATION LEADS
progress - Embrace new ideas

2 DATA-DRIVEN DECISIONS
- Use insights & analysis

3 CUSTOMER FOCUS REQUIRED
– Prioritize satisfaction & loyalty

4 STRATEGIC GROWTH
– Identify opportunities & expand markets

5 OPERATIONAL EFFICIENCY
– Streamline processes & resources

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